



Bare Metal to Proxmox VE Migration

Case Study



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Executive Summary

Lifebank Foundation Inc., a large non-profit organisation based in the Philippines, employs more than 5,000 staff and serves a significantly larger beneficiary and stakeholder base. To support its internal users, the organisation relies on a broad portfolio of digital services for collaboration, learning and business operations, including email, cloud storage, ERP, LMS, data science workloads, web meeting services and a comprehensive security and monitoring stack.

Key employee-facing workloads running on the platform include:

- Multiple mail servers for Lifebank's 5,000+ employees
- Multiple cloud storage servers for file sharing and collaboration
- ERP and LMS platforms supporting internal processes and training
- Data science and analytics workloads
- Web meeting and collaboration services
- A security and monitoring stack including NMS, SIEM, antivirus, log analysis, configuration backup and other business-critical solutions

Lifebank engaged Illumia Solutions Pvt. Ltd. (ISPL) to design and execute a progressive, open-source-based infrastructure modernisation programme. The engagement followed a two-stage evolution:

- Transition from fragmented bare-metal servers to a Proxmox-based hyperconverged infrastructure (HCI) with HDD/SSD-backed distributed storage, consolidating core services into a highly available virtualised platform.
- Subsequent deployment of a dedicated NVMe and SSD-based high-performance HCI cluster for the most demanding workloads, with selected critical applications migrated based on monitoring data and user experience.

ISPL also delivered ongoing AMC and managed services, covering end-to-end operations and security for the environment. The resulting landscape:

- Hosts more than 50 virtual machines across two production HCI clusters.
- Successfully migrated core services from dedicated bare-metal servers into virtual machines on the Proxmox HCI cluster, improving availability and manageability.
- Uses the original HDD/SSD-based cluster for suitable workloads and a newer NVMe cluster for high-IOPS applications.
- Achieves high-IOPS capacity increased by several times compared to the original cluster.
- Eliminates unplanned interruptions for IOPS-hungry applications, which dropped from frequent incidents to zero after the NVMe deployment.
- Implements daily backups to an on-site Proxmox Backup Server with replication to an off-site PBS node, reinforced by quarterly restore drills and regular VAPT/WAPT security assessments.
- Maintains a fully open-source stack, avoiding proprietary licensing lock-in and supporting long-term cost control.

Background Information

Before engaging ISPL, Lifebank's core services used by its employees ran on a set of dedicated bare-metal servers. Each major application – such as email, cloud storage and web meeting services – was hosted directly on its own physical server, with limited CPU, memory and HDD resources and no virtualization layer.

This pre-HCI environment was characterised by:

- Siloed bare-metal servers for individual applications
- Limited ability to pool compute and storage resources
- No built-in high availability across workloads
- Increasing operational complexity as more services were added

To address these constraints, Lifebank partnered with ISPL to move to a Proxmox-based HCI platform with distributed storage using HDD and SSD-based drives. The first HCI cluster:

- Centralised workloads previously spread across multiple bare-metal servers
- Enabled high availability and scale-out capacity via Proxmox VE and Ceph-based storage
- Introduced consistent backup using Proxmox Backup Server, with:
 - Daily backups to an on-site PBS node
 - Replication to an off-site PBS node for disaster recovery

This initial migration from bare-metal servers to the Proxmox HCI cluster gave Lifebank a unified, virtualised platform for its core services and a consistent backup and recovery framework.

Alongside the technical platform, ISPL established an AMC and managed services engagement that covered:

- HCI platform operations and networking for HCI
- VM commissioning and lifecycle management
- Monitoring, security and maintenance of the entire stack
- Administration of key applications such as mail, cloud and web meeting services
- Quarterly backup and restoration drills
- Periodic VAPT/WAPT security assessments

Over the next four years, Lifebank's usage and application portfolio expanded significantly as more internal teams and applications were moved onto the HCI platform. Especially mail, file and other I/O-intensive or analytical applications became increasingly demanding. The original HDD/SSD-based cluster continued to operate reliably, but:

- High-IOPS workloads began to experience performance limitations and occasional interruptions
- Risks to user experience during peak usage and to overall service continuity became more pronounced
- The organisation needed to accommodate future growth without disruptive rip-and-replace projects or major proprietary software investments

To protect both user experience and continuity of service while supporting future expansion, Lifebank and ISPL jointly decided to introduce a new NVMe and SSD-based high-performance HCI cluster and selectively migrate critical, high-IOPS workloads there, with weekend cutovers to avoid business-hours downtime.

Problem Statement

Lifebank's infrastructure evolution began with an urgent need to move away from fragmented bare-metal servers, and later had to accommodate rapid growth and high-IOPS workloads on top of the new HCI platform. The environment faced two successive but related sets of challenges.

1. Fragmented bare-metal servers and operational limitations

Before the initial Proxmox HCI deployment, Lifebank's core services ran directly on individual physical servers.

- Each major application – email, cloud storage, web meetings and other services – was hosted on its own dedicated bare-metal server.
- These servers had limited CPU, memory and HDD resources and could not share capacity efficiently.
- There was no virtualization layer, making it difficult to balance workloads or recover quickly from hardware failures.
- As more applications were added, operational complexity and management overhead increased steadily.

This bare-metal model made it hard to scale, maintain high availability and implement consistent backup and recovery across critical services.

2. Need for consolidation, high availability and consistent backup

As the organisation and its IT footprint grew, Lifebank needed a more modern platform to support its 5,000+ employees and internal users.

- Key services such as email, collaboration and learning had to be available consistently during business hours.
- The team needed a way to pool compute and storage resources to avoid stranded capacity and isolated bottlenecks.
- A standardised backup and recovery approach was required to protect all virtualised workloads, not just a subset.

These requirements led to the first major migration: from fragmented bare-metal servers to a Proxmox-based HCI cluster with Ceph storage and Proxmox Backup Server.

3. Business growth and emerging high-IOPS demands on the HCI cluster

After the initial migration, Lifebank continued to grow and consolidate more services onto the HCI platform.

- Additional workloads, including ERP, LMS, data science and expanded security/monitoring tools, were onboarded as virtual machines.
- Over several years, certain workloads – especially mail, file and analytical services – became increasingly I/O-intensive.
- Monitoring data and user feedback highlighted that some applications were starting to push the limits of the HDD/SSD-based cluster's storage performance.

While the original HCI cluster remained reliable overall, these high-IOPS workloads began to stretch its capabilities.

4. User experience and continuity risk under IOPS pressure

The growing IOPS demands translated directly into risk for both user experience and service continuity.

- IOPS-hungry workloads experienced latency and throughput constraints during peak usage, impacting responsiveness.
- Occasional performance-related interruptions were observed for these applications, even though the broader estate remained stable.
- Lifebank needed to protect day-to-day productivity for its employees and ensure that critical services such as email, collaboration and security monitoring did not degrade as workloads expanded.

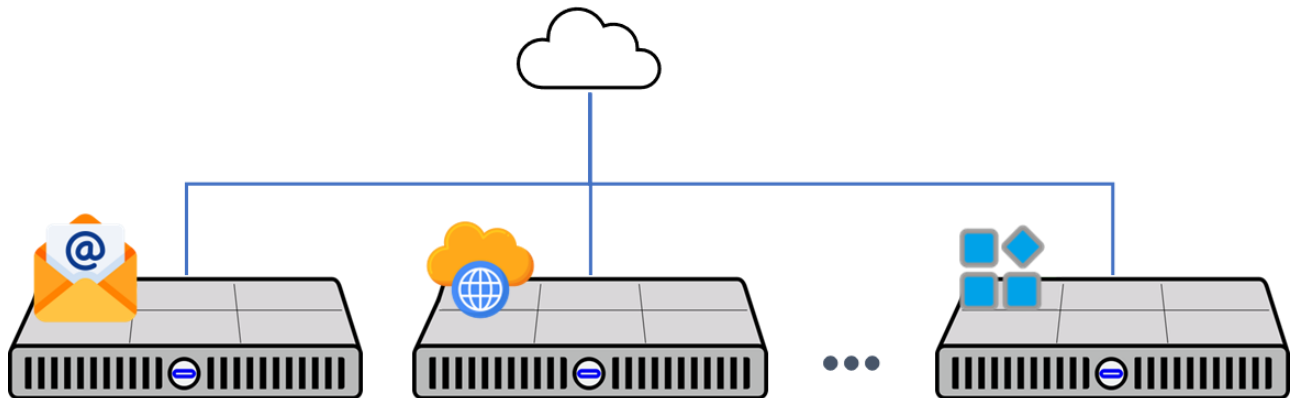
It was clear that performance had to be improved for a subset of workloads without disrupting the rest of the environment.

5. Requirement for a high-performance HCI tier without rip-and-replace

Lifebank wanted to address high-IOPS requirements by adding capacity, not discarding the original HCI investment.

- The HDD/SSD-based HCI cluster still had an important role for many workloads and should remain in production.
- A new, high-performance NVMe and SSD-based HCI cluster was needed to host the most demanding applications.
- Migrations had to be planned and executed during weekend windows to avoid downtime in business hours.
- The organisation also wanted to stay within its open-source strategy and avoid proprietary licensing lock-in.

These constraints defined the next stage of the journey: deployment of a high-performance NVMe HCI cluster and selective migration of high-IOPS workloads from the original HCI to the new platform.



Solution Proposed

ISPL proposed a progressive, two-stage solution that built on Lifebank's existing investments while adding high-performance capacity where it mattered most. The approach combined open-source HCI, NVMe-based storage, robust backup and managed services into a single operating model.

1. Open-source HCI as the core platform

The foundation of the design was a Proxmox-based hyperconverged infrastructure, standardising on open-source technologies across compute, storage and backup.

- Proxmox VE for virtualization, providing clustering, high availability and centralised management
- Ceph-based distributed storage using HDD and SSD data drives in the original cluster
- Proxmox Backup Server for image-level VM backups, with on-site and off-site PBS nodes
- An end-to-end open-source stack (mail, cloud, web meeting, PBS, PVE, NMS, SIEM and other components), keeping licensing exposure under control

The first stage of the solution was to migrate Lifebank's core services from dedicated bare-metal servers onto this Proxmox HCI platform, consolidating workloads and establishing a resilient, virtualised foundation.

2. Progressive HCI expansion instead of rip-and-replace

Rather than replacing the original HCI cluster, ISPL designed a progressive expansion model in which both clusters form a single logical estate.

- The existing HDD/SSD-based HCI cluster remained in production to host suitable workloads.
- A new, NVMe and SSD-based high-performance HCI cluster was introduced to handle high-IOPS workloads.
- Both clusters were integrated with shared backup, monitoring and logging infrastructure so they could be operated coherently as one environment.

This preserved prior investments while creating a clear path for performance-sensitive workloads.

3. Dedicated NVMe/SSD cluster for high-IOPS workloads

To relieve IOPS pressure on critical services and protect user experience, ISPL designed and deployed a new NVMe and SSD-based HCI cluster at Lifebank's central office.

- The cluster provided substantially higher IOPS and lower latency for demanding workloads.
- High-IOPS capacity increased by several times compared to the original HCI cluster.
- Selected critical applications were identified based on monitoring data and user experience, then migrated to the NVMe cluster to free resources on the original platform.

This targeted performance where it delivered the most impact while keeping the overall architecture simple.

4. Unified backup and data protection with PBS

The solution maintained and strengthened Lifebank's backup and DR posture using Proxmox Backup Server across both clusters.

- Daily backups of virtual machines to an on-site PBS node
- Replication of backup data to an off-site PBS node for disaster recovery
- Quarterly backup and restoration drills to validate recoverability and processes
- Integration of backup monitoring into the broader security and operations stack

This ensured that the move to a two-cluster HCI model did not compromise data protection.

5. Comprehensive AMC and managed services

To keep the environment healthy over time and reduce operational risk, ISPL provided ongoing AMC and managed services from the commissioning of the original HCI cluster.

- Operations and maintenance of both HCI clusters and the underlying networking for HCI
- VM commissioning and lifecycle management for new and existing workloads
- Monitoring, security and maintenance activities across the whole stack
- Administration of key applications such as mail, cloud and web meeting services
- Regular VAPT/WAPT security assessments and quarterly backup/restore drills

This operating model ensured that the technical design translated into sustained performance, reliability and security.

Implementation

ISPL executed the modernisation through a staged implementation, ensuring stability and continuity at each step.

Phase 1 – Transition from bare metal to HCI

The first phase focused on moving Lifebank away from fragmented bare-metal servers and onto a centralised HCI platform.

- Assessment of the existing bare-metal servers hosting email, cloud and web meeting services
- Design and commissioning of the initial Proxmox VE + Ceph-based HDD/SSD HCI cluster
- Migration of existing services from dedicated physical servers into virtual machines on the new cluster
- Validation with application owners to confirm performance, availability and functionality post-migration

This phase completed the migration away from fragmented bare-metal servers and moved Lifebank's core services onto the Proxmox HCI platform.

Phase 2 – Establishment of unified backup and operations

Once the bare-metal to HCI migration was complete and stable, ISPL standardised backup and operations on top of the new platform.

- Deployment and integration of Proxmox Backup Server as the backbone of VM protection
- Configuration of daily backups to an on-site PBS node and replication to an off-site PBS node
- Set-up of monitoring, logging and security tools across the HCI stack
- Onboarding of AMC and managed services, including routine maintenance and security assessments

This phase turned the new platform into an operationally robust environment.

Phase 3 – Performance monitoring and identification of high-IOPS workloads

Over several years, Lifebank's application landscape expanded and ISPL continuously monitored performance and utilisation.

- Analysis of resource usage and I/O patterns across VMs
- Identification of workloads that were IOPS-intensive or latency-sensitive, such as specific mail, file and analytical applications
- Correlation of monitoring data with user feedback to confirm which services were most impacted by storage performance limits

This informed the design of the next expansion step.

Phase 4 – Deployment of NVMe/SSD high-performance HCI cluster

In response to observed IOPS starvation and growth, ISPL deployed a new high-performance HCI cluster based on NVMe and SSD storage.

- Design and commissioning of the NVMe/SSD cluster at Lifebank's central office
- Integration with existing backup, logging and monitoring systems so both clusters could be managed as one estate
- Configuration of high-speed, dedicated storage networking to ensure low-latency access to NVMe storage

The new cluster was validated before any production workloads were moved.

Phase 5 – Migration of selected high-IOPS workloads

With the new cluster ready, ISPL executed controlled migrations of critical, high-IOPS workloads.

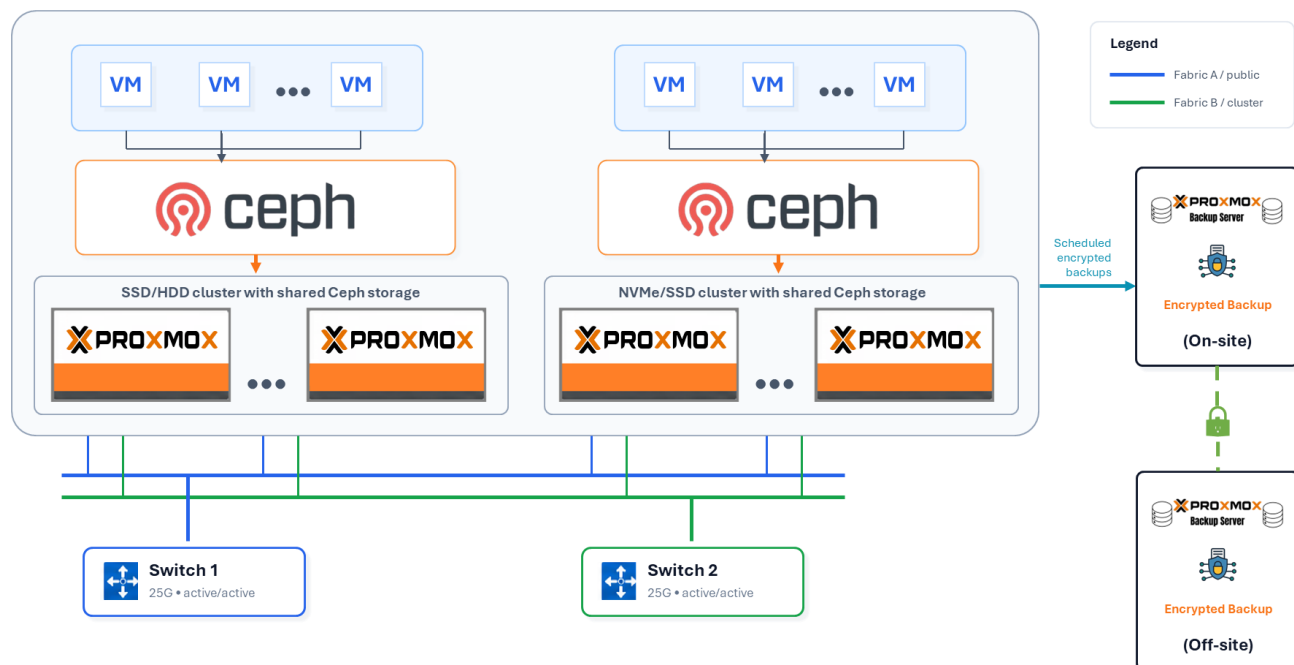
- Selection of target VMs based on monitoring data and user experience impact
- Planning and execution of cutovers during weekend windows to avoid business-hours downtime
- Migration of selected VMs from the original HCI cluster to the NVMe/SSD cluster
- Post-migration validation with application owners to confirm performance improvements and stability

This phase completed the targeted migration of high-IOPS workloads from the original HDD/SSD HCI cluster to the new NVMe/SSD cluster, increasing high-IOPS capacity by several times and eliminating unplanned interruptions for IOPS-hungry applications.

Phase 6 – Continuous operations, optimisation and assurance

Following the migrations, ISPL continued to operate and refine the environment under the AMC and managed services engagement.

- Ongoing monitoring of both clusters and key applications
- Regular capacity reviews to plan for future growth
- Quarterly backup/restore drills and periodic VAPT/WAPT assessments to maintain security and recoverability
- Continuous tuning of workloads across the two clusters to balance performance, resilience and resource utilisation



Conclusion

Through the staged deployment of an open-source HCI platform and a dedicated NVMe/SSD cluster, supported by PBS and managed services, Lifebank realised:

- A consolidated, virtualised infrastructure hosting more than 50 VMs across two production HCI clusters.
- Successful migration of core services from dedicated bare-metal servers into the initial Proxmox HCI cluster, establishing a single, resilient platform for email, collaboration and other key applications.
- High-IOPS capacity increased by several times, with unplanned interruptions for IOPS-hungry applications dropping from frequent incidents to zero after the NVMe deployment.
- A robust data protection posture via daily backups to an on-site PBS and replication to an off-site PBS node, validated by quarterly restore drills.
- End-to-end managed operations covering HCI, networking for HCI, VM commissioning, monitoring, security and key application administration.
- Continued use of an open-source stack for mail, collaboration, web meetings, backup, virtualisation, monitoring and security, avoiding proprietary licensing lock-in and supporting long-term cost savings.

Broader significance

For Lifebank, the evolution from bare-metal servers to a dual-cluster HCI environment strengthened both user experience and continuity of service for its 5,000+ employees and wider beneficiary base. The combination of open-source HCI, targeted NVMe expansion, robust backups and structured managed services allowed the organisation to scale performance where needed, protect critical workloads and maintain predictable operating costs.

This case provides a reference model for non-profit and microfinance organisations seeking to modernise their infrastructure: start by consolidating onto an open-source HCI core, then progressively add high-performance capacity and operational maturity as workloads and expectations grow, rather than attempting disruptive rip-and-replace projects.

Hardware Infrastructure

